Contents

Preface

These notes cover the Analysis 1 course taught during Aug-Nov 2026 at CMI by Prof. Karandikar. Throughout the course the emphasis will be on rigorous proofs of the fundamental theorems used in Calculus. Construction of reals, and metric spaces will *not* be covered.

Timings

$2:00 - 3:15$pm Tuesday, Thursday and $11:50 - 12:05$pm on Wednesday. Shared tutorials with Algebra every Monday and Friday at 2pm.

Textbook

Understanding Analysis by Stephen Abbott is a good reference for this course. He recommends *Principles of Mathematical Analysis* by Walter Rudin, which I don't recommend as a first text.

Syllabus

1. Axioms of the real number system without construction, applications of the least-upper-bound property, Archimedean principle, existence of nth roots of positive real numbers, a^x for $a > 0$ and $x > 0$, cardinality, countability of rational numbers, uncountability of real numbers.
2. Convergence of sequences, uniqueness of limit, Sandwich lemma, exam- ples, monotonic sequences, subsequences, Bolzano-Weierstrass theorem, lim sup and lim inf, Cauchy sequences, completeness of $\mathbb{R}$.
3. Infinite series, convergence and divergence, absolute convergence, condi- tional convergence, convergence/divergence of $\sum_{k=1}^{\infty} 1/k^p$, comparison test, root test, ratio test, Leibniz test.
4. Complex numbers, power series, radius of convergence of power series.
5. Continuous functions on intervals of $\mathbb{R}$, intermediate value theorem, boundedness of continuous functions on closed and bounded intervals. Uniform continuity, uniform convergence, uniform limit of continuous functions is continuous, counter example for point wise convergence.
6. Differentiation, mean value theorem, Taylor's theorem, application of Taylor's theorem to maxima and minima, L'Hôpital rule.
7. Construction of ez using power series, proof of the periodicity of sin and cos.
8. Riemann Integration: Riemann integrals, Riemann integrability of continu- ous functions, fundamental theorem of calculus. Improper integrals

Grading

Homework will *not* be graded. The grading will be distributed between quizzes, midsem and finals.

Goal

These notes are intended for an audience that is familiar with basic highschool mathematics, in particular the notions of functions, basic set theory, etc. Beyond this, I try not to assume any prior knowledge. The goal is to produce a more comprehensive, if slightly verbose, introduction to analysis, that tries not to omit details that otherwise might be left out, and tries to motivate the ideas presented through intuitive notions and visualizations.

For this reason, most exercises have a solution provided. I usually at least provide a sketch, unless the exercise is very simple. I recommend that you try to solve the exercise yourself for a while before looking at the solution. I have taken some liberty in rearranging the questions from the course, so that some maybe a part of the main text itself, while some are provided as exercises.

In particular, I hope this will serve as a reference for any of my classmates taking this course, who have no prior background in analysis. If you find any errors, or have any suggestions please mail me at lunalgia@pm.me.

1 Sequences

1.1 Preliminaries

Our aim is here to rigorously deal with the various results you have probably already encountered in calculus. While we will be *rigorous*, we won't be *abstract*. You have probably already encountered limits of functions. One way to think about such limits is *sequences*. In highschool mathematics, you might have encountered "intuitive" notions of sequences:

Sequence: a sequence is a list of numbers, e.g. $a_1, a_2, a_3, \ldots$

Let us work with this for now. Consider the following limit:

$$\lim_{x \to 0} \sin(x) = 0.$$

One way to think about it is by getting closer and closer to 0 in a continuous manner. However, an equivalent, and maybe more powerful way is to think about a list of numbers that grows closer and closer to 0. So, let us plot $\sin(x)$ for $x = x_1, x_2, x_3, \ldots$ where $x_n \to 0$. This allows us to see that $\sin(x)$ approaches 0 as x grows closer to 0.

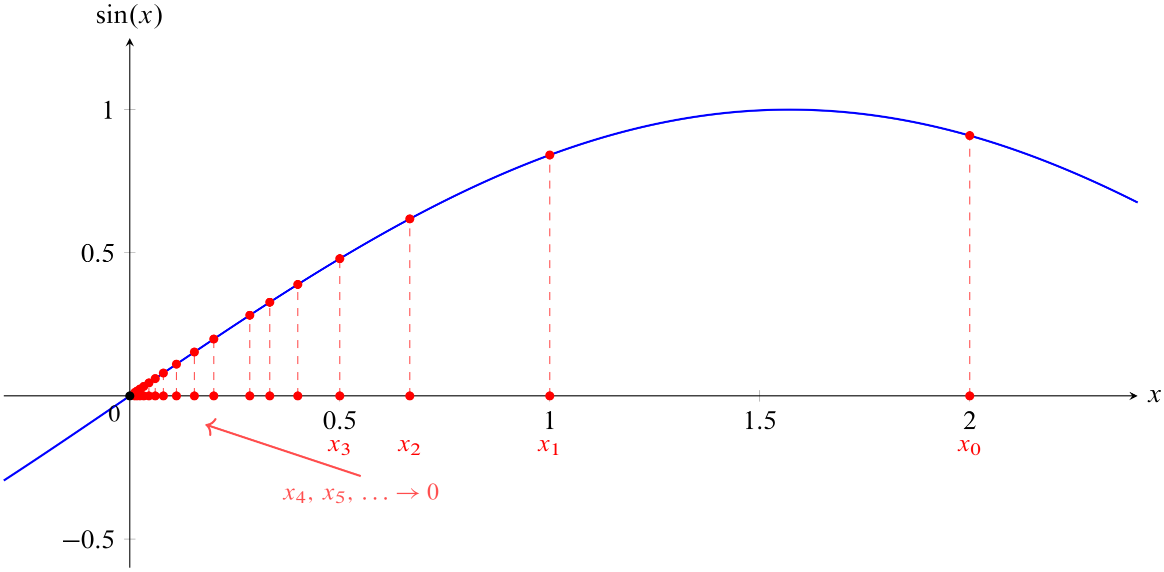

Figure 1.1. A sequence $x_i \to 0$ and the corresponding values $\sin(x_i) \to 0$.

This is primarily the reason we work with sequences. Although it is easy to imagine the limit in the other way too, it is often helpful to work with a discrete set of points instead.

However the definition of sequence we laid out hardly suffices as a good definition, for one, we have postponed the burden of definition from sequence to a list of numbers. Let us give a more formal definition.

Definition 1.1

A *sequence* is a function $a : \mathbb{N} \to \mathbb{R}$, such that

$$a_n = a(n)$$

for all $n \in \mathbb{N}$ is the n-th term of the sequence.

I will adopt the convention that $\mathbb{N}$ does not include 0. The notions of functions and other primitive things will be assumed. Sequences will be denoted by $(a_n)_{n=1}^{\infty}$ or simply (a_n), for reasons that will become clear as we go.

Example 1.1

Some examples from highschool mathematics include the sequences defined by $a_n = ar^n$, and $a_n = a + nd$. These sequences are *geometric* and *arithmetic*, respectively.

For this course we will assume the properties and existence of $\mathbb{R}$. In more rigorous analysis courses, you will learn about the *construction* of $\mathbb{R}$, but we will not go into detail here. The properties of $\mathbb{R}$ are listed below.

- For all $a, b \in \mathbb{R}$, $a + b, a - b, a \cdot b, a/b$ are defined (for the last, $b \neq 0$).
- For all $a, b \in \mathbb{R}$, either $a < b$, $a = b$, or $a > b$.

You might realise that these properties are also the properties that $\mathbb{Q}$ has. What, then, distinguishes $\mathbb{Q}$ from $\mathbb{R}$? Clearly they're not the same, we've all seen proofs that $\sqrt{2}$ is in fact not rational, but we also happen to "know" that $\sqrt{2}$ is a real number. There is precisely one property that distinguishes $\mathbb{Q}$ from $\mathbb{R}$, and it happens to unfortunately be a little more obtuse than the above, rather obvious ones.

The property we speak of is the *Least Upper Bound Property*, but before stating it, we need to define what an *upper bound* is.

Definition 1.2

A *least upper bound* of a set S is a real number b such that $b \geq a$ for all $a \in S$.

We now must define what it means to have a *least upper bound* for a set S:

Definition 1.3

A set S has a *least upper bound* if there exists a real number b such that b is an upper bound for S, and if a is an upper bound for S, then $b \geq a$. We write $b = \sup S$.

Now we can actually state this mysterious property of the reals:

If $S \subset \mathbb{R}$ has an upper bound b, then S has a least upper bound b.

There is a similar property for lower bounds, that I will leave for you to prove, using the least upper bound property.

Definition 1.4

A set S has a *least lower bound* if there exists a real number b such that b is a lower bound for S, and if a is a lower bound for S, then $b \leq a$. We write $b = \inf S$.

One final property that we require is the archimedean property. This seems like a really obvious result, so much so that you might be surprised that it is even necessary. However as we will continue to do in the course, we will not take it as a given and actually prove it.

Theorem 1.1

Archimedean Property

For any real number a, there exists a positive integer N such that $N > a$.

Proof. Suppose no such N exists. Then for every positive integer N, $N \leq a$, so a is an upper bound of $\mathbb{N} \subset \mathbb{R}$. By the Least Upper Bound Property, there must exist a least upper bound b of $\mathbb{N} \subset \mathbb{R}$.

Note that $b - 1$ must be an upper bound of $\mathbb{N}$ as well, because if $b - 1 < n \in \mathbb{N}$, then $b < n + 1 \in \mathbb{N}$, which cannot be true since b is an upper bound of $\mathbb{N}$. However, if $b - 1$ is an upper bound of $\mathbb{N}$, then $b - 1 > b$ by the definition of Least Upper Bound, which cannot be true! Hence some $N \in \mathbb{N}$ must exist such that $N > a$. □

Exercises

1.1.1. Prove that if a set has a lower bound, then it has a greatest lower bound.

1.2 Convergent Sequences

Although just about anything can be defined as a sequence—for example the oscillating sequence defined by $a_n = (-1)^n$—our interest resides in a more limited set of "nice" sequences. Consider for example the sequence defined by $a_n = 1 + 1/n$. As we go to farther values of n, this sequence grows closer and closer to the value 1. This notion of growing closer and closer to a value is captured by the concept of convergence (which is equivalent to the idea of a limit).

Definition 1.5

A sequence (a_n) is *convergent*, and is said to converge to $\limsup a_i$, called the *limit* of the sequence, if for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $|a_n - \limsup a_i| < \varepsilon$ for all $n \geq N$, where $\limsup a_i$ is the limit of the sequence.

We write $\lim_{n\to\infty} a_n = \limsup a_i$ to denote that the sequence (a_n) converges to $\limsup a_i$. If the definition looks a little obtuse, then do not worry. At first it might seem terribly confusing, but this actually just captures the notion we just mentioned, the point is that by going to large enough values, we can get arbitrarily close to the value it converges to.

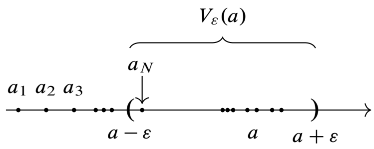

Figure 1.2. A sequence that converges to a

Remark 1.2.1. There is a *topological* way to think about this. Unfortunately the course won't go into this in detail, but something useful is the notion of ε neighbourhood of a point. This is simply the set of all points within ε distance of the given point. So how is it different from the definition we just gave?

The notion of distance is not just limited to $\mathbb{R}$—while we do have $|x - a|$ as the distance of x from a, we can also find out the distance between two points in for example $\mathbb{R}^2$, using what is called the *metric*. For now these are just words without meaning, but perhaps the idea of convergence by being able to find a point in the sequence for any $\epsilon > 0$, such that all terms after it are within the ε neighbourhood, V_ε of that point.

A good imperative to have when defining something is one, to actually see if this object exists, and two to see that it is well defined. When we write down a symbol like $\lim_{n\to\infty} a_n$, we must check if this is actually a unique value, that is there is only value that the sequence converges to. Otherwise our definition would be ambiguous, the symbol would mean too many things. Therefore, let us do a little proposition.

Proposition. *If the sequence* (a_n) *converges to* $\limsup a_i$, *and* M, *then* $\limsup a_i = M$.

Proof. Suppose the sequence (a_n) converges to $\limsup a_i$, and M. Then, by the definition of convergence, for any $\varepsilon > 0$, there exists N_1 such that for all $n \geq N_1$, $|a_n - \limsup a_i| < \varepsilon$. Similarly, there exists N_2 such that for all $n \geq N_2$, $|a_n - M| < \varepsilon$.

A straightforward way to proceed is to add up the two inequalities. But first you must remember that the n that satisfied $n > N_1$ does not necessarily satisfy $n > N_2$. How do we deal with this? Pick any $n > \max(N_1, N_2)$, so that now a_n satisfies both the inequalities.

$$|a_n - \limsup a_i| + |a_n - M| = |a_n - \limsup a_i| + |M - a_n| < 2\varepsilon,$$

and now we use one of the most important inequalities in these epsilon proofs, the triangle inequality:

$$2\varepsilon > |a_n - \limsup a_i| + |a_n - M| \geq |a_n - \limsup a_i| + |M - a_n| = |M - \limsup a_i|.$$

But remember that we could choose *any* epsilon here. Now there are two ways to proceed. One, by choosing the initial epsilon to be $\varepsilon'/2$, we have that for any $\varepsilon' > 0$,

$$\varepsilon' > |M - \limsup a_i|$$

but that must mean $M - \limsup a_i = 0$! (convince yourself of it, by arguing formally using $M - \limsup a_i = r \neq 0$).

A more clever proof would be to use $\varepsilon = |M - \limsup a_i|/2$, assuming for contradiction that $M \neq \limsup a_i$, so that we would get:

$$|M - \limsup a_i| < 2\varepsilon = |M - \limsup a_i|$$

which cannot be! The idea behind this is to think about close a_n must be to $\limsup a_i$ and M, and achieve a contradiction by thinking about the distance between the two values. □

Now we know that our symbols are well defined. But I have yet to show you an example of a sequence that converges to some value using the definition we adopted. Let us now do that.

Example 1.2

The sequence $a_n = \dfrac{1}{n}$ converges to 0.

Before writing the proof, let us think about what we need. The definition requires: for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $|a_n - 0| < \varepsilon$ for all $n > N$. The quantifier, for any, matters here. ε is given to us first (adversarially, we may imagine), and only afterward do we get to choose N; the choice of N is therefore allowed to depend on ε.

Fix an $\varepsilon > 0$, how large must n be to guarantee $|1/n - 0| < \varepsilon$? Since $1/n > 0$, this inequality is equivalent to $n > 1/\varepsilon$. This tells us exactly how to choose N: any natural number $N \geq 1/\varepsilon$ will do. We now write this up formally.

Proof. Let $\varepsilon > 0$. By the Archimedean property, there exists $N \in \mathbb{N}$ with $N > 1/\varepsilon$. We claim this N works. Let $n > N$. Then

$$n > N > \frac{1}{\varepsilon} \quad \Longrightarrow \quad \frac{1}{n} < \varepsilon,$$

and since $n \geq 1$ we have $1/n > 0$, so

$$|a_n - 0| = \frac{1}{n} < \varepsilon.$$

As $\varepsilon > 0$ was arbitrary, this shows $a_n \to 0$. ■ □

Example 1.3

The sequence $a_n = \dfrac{2n}{3n+1}$ converges to $\dfrac{2}{3}$.

As before, we first manipulate $|a_n - \limsup a_i|$ algebraically until the dependence on n becomes transparent:

$$\left|\frac{2n}{3n+1} - \frac{2}{3}\right| = \left|\frac{3(2n) - 2(3n+1)}{3(3n+1)}\right| = \left|\frac{-2}{3(3n+1)}\right| = \frac{2}{3(3n+1)}.$$

This is a decreasing function of n, so making it small is just a matter of making n large. We want

$$\frac{2}{3(3n+1)} < \varepsilon \quad\Longleftrightarrow\quad 3n+1 > \frac{2}{3\varepsilon} \quad\Longleftarrow\quad n > \frac{2}{9\varepsilon}.$$

(We used $\Longleftarrow$ rather than $\Longleftrightarrow$ in the last step purely for convenience — we don't need the sharpest possible N, only *some* N that works, so we are free to bound things generously.) This suggests taking N to be any natural number exceeding $2/(9\varepsilon)$.

Proof. Let $\varepsilon > 0$, and choose $N \in \mathbb{N}$ with $N > \dfrac{2}{9\varepsilon}$ (possible by the Archimedean property). Let $n > N$. Then $n > 2/(9\varepsilon)$, so $9\varepsilon n > 2$, and hence $3(3n+1) = 9n + 3 > 9n > \dfrac{2}{\varepsilon}$. Rearranging,

$$\frac{2}{3(3n+1)} < \varepsilon.$$

Combined with the algebraic identity from the discussion above,

$$\left|\frac{2n}{3n+1} - \frac{2}{3}\right| = \frac{2}{3(3n+1)} < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $a_n \to 2/3$. ■

You might've realised that this is often quite cumbersome to work with. For example, what about proving $a_n = 1/n + 2n/(3n+1)$ converged to $2/3$? Try doing this yourself, but be warned, this will be annoying to deal with.

We can simplify many matters here by proving some useful properties of limits of sequences. These are set up as exercises to be done at the end of this section. They're guided of course, since picking the right epsilon is a skill that requires learning which I will not presume.

Remark 1.2.2. A subsequent definition that we can make is to formalise the notion of divergence. A sequence (a_n) is said to *diverge* if there does not exist a value that the sequence converges to.

However it serves to distinguist between two different notions of divergence. Consider the sequences $a_n = (-1)^n$ and $b_n = n$. Both sequences diverge, but the first one has no well defined behaviour as we grow larger, while the larger gets arbitrarily large.

To formalise this distinction, we often work in the *extended real numbers* defined by

$\mathbb{R} \cup \{\infty, -\infty\}$, where ∞ and $-\infty$ represent the notions of positive and negative infinity, respectively. We define a few properties:

- For any $a \in \mathbb{R}$, $a \leq \infty$ and $a \geq -\infty$.
- For any $a \in \mathbb{R}$, $a + \infty = \infty$ and $-\infty + a = -\infty$.

Now we are able to extend the notion of convergence:

- A sequence (a_n) converges to ∞ if for any $M \in \mathbb{R}$, there exists a N such that $a_n > M$ for all $n \geq N$.
- A sequence (a_n) converges to $-\infty$ if for any $M \in \mathbb{R}$, there exists a N such that $a_n < M$ for all $n \geq N$.

Exercises

We will prove the following properties of limits of sequences:

- If $\lim_{n\to\infty} a_n = a$, and $\lim_{n\to\infty} b_n = b$, and c_n is the sequence defined by $c_n = a_n + b_n$, then $\lim_{n\to\infty} c_n = a + b$.
- If $\lim_{n\to\infty} a_n = a$, and $\lim_{n\to\infty} b_n = b$, and c_n is the sequence defined by $c_n = a_n \cdot b_n$, then $\lim_{n\to\infty} c_n = a \cdot b$.
- If $\lim_{n\to\infty} a_n = a$, and $\lim_{n\to\infty} b_n = b \neq 0$, and c_n is the sequence defined by $c_n = a_n/b_n$ when $b_n \neq 0$, and 0 when $b_n = 0$, then $\lim_{n\to\infty} c_n = a/b$.

1.2.2. Prove that if

$$|x - x_0| < \frac{\varepsilon}{2} \quad \text{and} \quad |y - y_0| < \frac{\varepsilon}{2},$$

then

$$|(x + y) - (x_0 + y_0)| < \varepsilon,$$
$$|(x - y) - (x_0 - y_0)| < \varepsilon.$$

1.2.3. Prove that if

$$|x - x_0| < \min\left(\frac{\varepsilon}{2(|y_0| + 1)}, 1\right) \quad \text{and} \quad |y - y_0| < \frac{\varepsilon}{2(|x_0| + 1)},$$

then $|xy - x_0y_0| < \varepsilon$.

(The notation "min" is self evident, the first inequality in the hypothesis just means that

$$|x - x_0| < \frac{\varepsilon}{2(|y_0| + 1)} \quad \text{and} \quad |x - x_0| < 1;$$

at one point in the argument you will need the first inequality, and at another point you will need the second. One more word of advice: since the hypotheses only provide information about $x - x_0$ and $y - y_0$, it is almost a foregone conclusion that the proof will depend upon writing $xy - x_0 y_0$ in a way that involves $x - x_0$ and $y - y_0$.)

1.2.4. Prove that if $y_0 \neq 0$ and

$$|y - y_0| < \min\left(\frac{|y_0|}{2}, \frac{\varepsilon |y_0|^2}{2}\right),$$

then $y \neq 0$ and

$$\left|\frac{1}{y} - \frac{1}{y_0}\right| < \varepsilon.$$

1.2.5. Now prove the three propositions we stated at the beginning.

1.2.6. We call a sequence bounded iff there exists a number M such that $|x| \leq M$ for all x in the sequence. We call a sequence monotonically increasing iff for all $n \leq m$, $x_n \leq x_m$.

Suppose that (a_n) is a monotonically increasing bounded sequence. Prove that it converges.

Hint: Let $B = \{a_n : n \in \mathbb{N}\}$, the set of all values of a_n. Note that this is bounded. Then consider the least upper bound of B. How does this relate to the convergence of (a_n)?

1.2.7. Show that every convergent sequence is bounded.

1.3 Cauchy Sequences

Sometimes we know that a sequence does converge, but using the above definition of convergence makes it hard to prove that it does. A more powerful definition is the Cauchy sequence definition.

Let us go through an example. To estimate square roots, a method that has been used since ancient times is the Babylonian method, which starts with a guess and repeatedly averages it with 37 divided by itself. Let us define a sequence using it, with initial value 6 (since it is close to the square root of 37):

$$x_1 = 6, \qquad x_{n+1} = \frac{1}{2}\left(x_n + \frac{37}{x_n}\right)$$

n	x_n	$\lvert x_n - x_{n-1}\rvert$
1	6.000000	—
2	6.083333	0.083333
3	6.082763	0.000571
4	6.082763	0.000000
5	6.082763	0.000000

TABLE 1.1 . The first few terms of the Babylonian square root sequence. x_4 and x_5 are so close that they have the same values upto 6 decimal places.

I hope it is clear to you that this sequence converges to whatever the value of $\sqrt{37}$ is. If you're not convinced, write down a few more values of the sequence, and use whatever value of $\sqrt{37}$ your calculator gives you. But how do you go about proving this? It seems very inelegant to have to rely on a calculator to prove that a sequence converges, nevermind if it even suffices as a proof [1].

But I have deliberately written down the third column of the table here, so that you can see that the difference between the terms of the sequence decreases as n increases. Roughly speaking, then the sequence should also converge, as the following terms don't deviate too much. This is infact the idea behind Cauchy sequences, which allow to use restate convergence.

1. It can be made rigorous by proving the decimal expansion of $\sqrt{37}$ to some number of decimal places. Then you could try to show that a term of the sequence has the same value as the decimal expansion of $\sqrt{37}$ up to that point, so that their difference is less than 10^{-m} for some m. I hope that this comes across as really cumbersome to you.

Definition 1.6

A Cauchy sequence is a sequence (a_n) such that for any $\varepsilon > 0$, there exists an N such that $|a_n - a_m| < \varepsilon$ for all $n, m \geq N$.

Let us prove the proposition we just discussed, of how Cauchy sequences relate to convergence.

Theorem 1.2

A sequence is cauchy if and only if it is convergent.

Unfortunately the proof of this will have to wait for a while. One direction of it is not hard, I will leave it up to you to show that a convergent sequence is Cauchy. However, for the other direction we'll need to cover a few more definitions. It is perhaps clear that we would need to do this, because to prove something is convergent, you would want a value that it converges to. For an arbitrary sequence, it seems outside our abilities to be able to choose something like that for now.

Exercises

1.3.8. Show that a convergent sequence is Cauchy.

1.3.9. Show that any Cauchy sequence is bounded

1.4 Limit Points

Although a sequence might not have a limit, it might still "cluster" around a value infinitely often. So we could define a new sequence, consisting of such terms in the sequence, and it would in fact converge to such a point. This is the idea behind *limit points*.

Definition 1.7

We call θ a limit point of a sequence, if for all $\varepsilon > 0$, and $M \geq 1$, there exists $N \geq M$ such that $|a_N - \theta| < \varepsilon$.

Be aware of the distinction between the definition of a limit point and a limit! Here we are just claiming we can find an arbitrarily large point close to θ, not that every point after that is also close to θ. So for example, a sequence like $a_n = (-1)^n$ has -1 as a limit point, but it does not converge to -1.

The idea of defining a new sequence derived from the original also has a name, in particular, we define a *subsequence* of the original sequence.

Definition 1.8

A sequence (b_m) is called a *subsequence* of the original sequence a_n if there exists a sequence $1 \leq n_1 < n_2 < \cdots < n_m < \dots$ such that $b_m = a_{n_m}$ for all m.

Therefore, we should be able to find a subsequence that converges to any limit point. Since it is a good idea to learn how to define a subsequence with specific properties, we will prove this result.

Theorem 1.3

If a sequence (a_n) has a limit point θ, then there exists a subsequence (b_m) that converges to θ.

Proof. We want to ensure that we can get arbitrarily close to θ by going far enough into the sequence. How do we do this? Well, the first thing is to utilise the definition of the limit point. We know that we can find some point, n_1 that is arbitrarily close to θ. But this is for a particular value of ε, and sure we could keep choosing $n_2 > n_1$ to get a term that is similarly close to θ, but that is hardly sufficient.

The idea is to get stricter and stricter bounds. Define a subsequence as follows, let n_1 be such that $\left|a_{n_1} - \theta\right| < 1$. Then let $n_2 > n_1$ be such that $\left|a_{n_2} - \theta\right| < 1/2$, and so on, that is $n_m > n_{m-1}$ be such that $|a_{n_m} - \theta| < 1/m$. Define $b_m = a_{n_m}$. Let us prove that this indeed works.

Consider an epsilon, we must find a $N \in \mathbb{N}$ such that for all $n > N$, $|b_n - \theta| < \varepsilon$. We know that $|b_n - \theta| < 1/n$, so it suffices to choose N such that $1/N < \varepsilon$, i.e. $N > 1/\varepsilon$, which we can indeed do by the Archimedean property, so we are done. $\square$

> **Remark 1.4.1.** This construction really reveals the power of sequences. You can always work with functions by default, but using sequences allow you to *inductively* build up a desired sequence. This is not something you can do with functions, as might be obvious to you.

Can a sequence cluster around any point? Clearly not if it is bounded at least. But there is actual a stricter sense in which it can't cluster around any point—for bounded sequences—using the idea of lim sup and lim inf.

Definition 1.9

> For a sequence (a_n), the lim sup is the limit of the sequence of upper bounds of the *tails of the sequence*, $\limsup a_n := \lim_{n\to\infty} \sup\{a_m \mid m \geq n\}$, and the lim inf is the limit of the sequence of lower bounds of the *tails of the sequence*, $\liminf a_n := \lim_{n\to\infty} \inf\{a_m \mid m \geq n\}$.

Limit points, lim sup, and lim inf of a bounded sequence

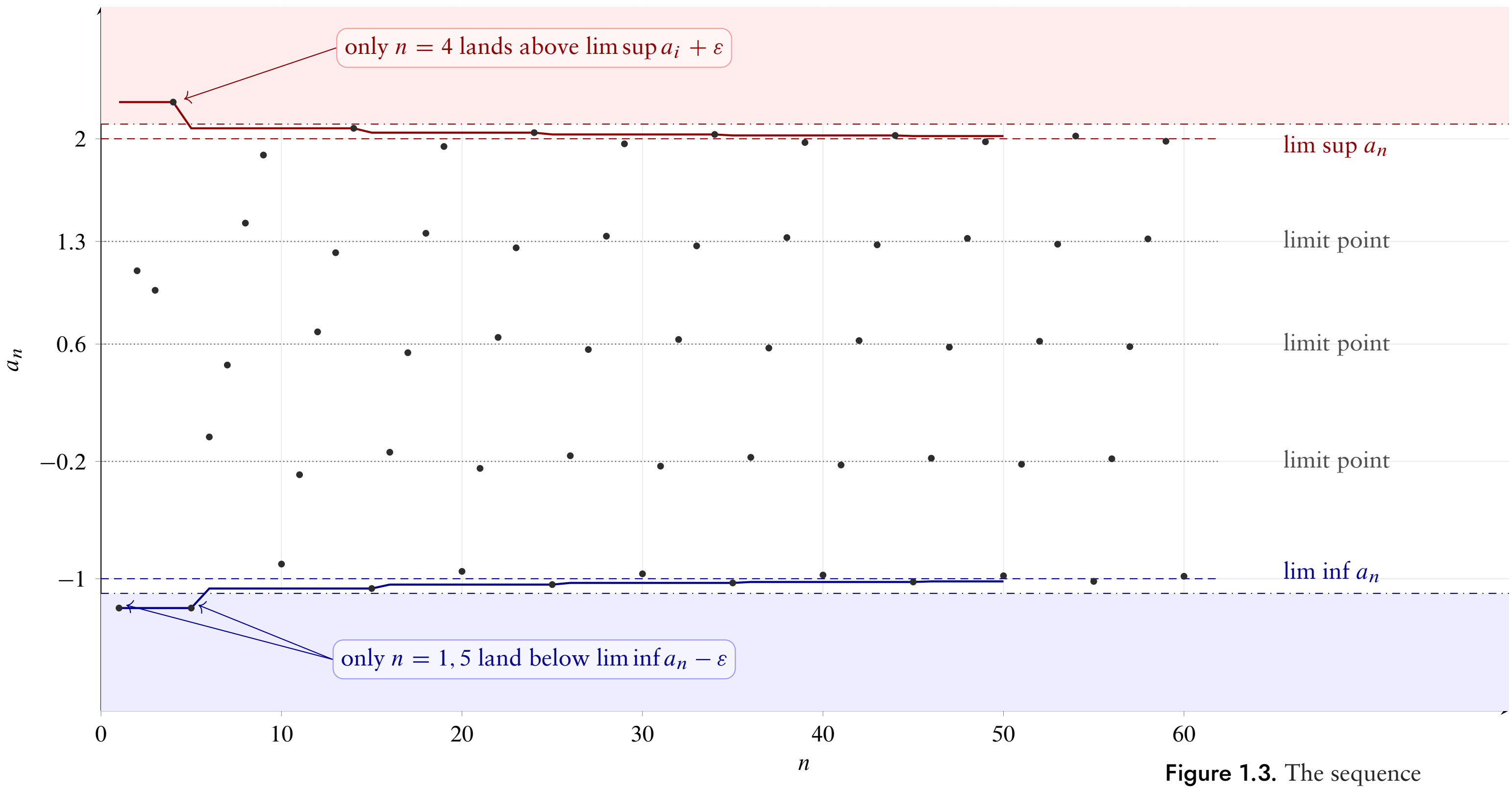

Figure 1.3. The sequence $a_n = c_{n \bmod 5} + (-1)^n/n$ with $c_0, \ldots, c_4 = -1, -0.2, 0.6, 1.3, 2$. One can see that five subsequences cluster at five distinct values. Red/blue curves are $\sup_{n\geq N} a_n$ and $\inf_{n\geq N} a_n$, which converge to $\limsup a_i$ and $\liminf a_n$. For any $\varepsilon > 0$ (here $\varepsilon = 0.1$), only finitely many terms lie above $\limsup a_i + \varepsilon$ or below $\liminf a_n - \varepsilon$.

We will show that these exist for convergent sequences (more generally, for bounded sequences), and these are the tools that will finally allow us to show that all Cauchy sequences are convergent. But first let us discuss about the idea of clustering I mentioned—what the lim sup and lim inf are really doing. As I mentioned, they are the greatest and smallest limit points. That is, the function clusters around a value greater than the lim sup and smaller than the lim inf only finitely many times.

Before we prove Theorem 1.2, let us first discuss this a little. We want to show that every bounded sequence has lim sup and lim inf as limit points. But to do that at all, we must show these actually exist, based on our definition.

Theorem 1.4

If (a_n) is a bounded sequence, then the sequence defined by $t_n = \sup\{a_m \mid m \geq n\}$ and $s_n = \inf\{a_m \mid m \geq n\}$ is convergent, and $\limsup a_n = \lim_{n\to\infty} t_n$ and $\liminf a_n = \lim_{n\to\infty} s_n$.

Proof. I will only do one of these, and leave the other for you to fill in.

We will do this in a silly manner, first showing that (t_n) is bounded, and then that is monotonically decreasing[2]. This will allow us to use Exercise 1.2.6. to conclude that (t_n) is convergent.

2. The definition is analogous to that of monotonically increasing sequence, i.e $t_n \geq t_m$ for all $m \geq n$.

The first is easy to conclude from Exercise 1.3.9.. Since (a_n) is bounded, there exists M such that $|a_n| \leq M$ for all n. Then, $t_n \geq a_n \geq -M$ for all n. Also, since M is an upper bound for $\{a_m \mid m \geq n\}$, $t_n \leq M$ for all n. Therefore, (t_n) is also bounded.

Now for the latter, we note that t_n is the least upper bound of $\{a_m \mid m \geq n\}$, and t_{n+1} is the least upper bound of $\{a_m \mid m \geq n+1\}$. If $t_n < t_{n+1}$, then t_{n+1} is not the least upper bound of $\{a_m \mid m \geq n+1\}$, $t_n \geq a_m$ for all $m \geq n+1$, so it is an upper bound, and $t_n < t_{n+1}$, which is a contradiction. Therefore, we must have that $t_n \geq t_{n+1}$, and so (t_n) is monotonically decreasing.

Note that the sup exists because the sequence is bounded, so the tail has an upper bound.

To prove that it is convergent, we will use Exercise 1.2.6. with a bit of trickery. It is easy to see that $t'_n = -t_n$ is bounded and monotonically increasing, so it must be convergent. Also, $-\mathbb{1}_n = -1$ is also convergent (it converges to -1), so $t_n = -\mathbb{1}_n \cdot t'_n$ must also be convergent by Exercise 1.2.5.. By definition, we write $\limsup a_n = \lim_{n\to\infty} t_n$. □

Now let us consider the result we talked about:

Theorem 1.5

If (a_n) is a bounded sequence, then $\limsup a_n$ is a limit point of (a_n).

Proof. We must show that for all $\varepsilon > 0$ and $M \geq 1$, there exists $N \geq M$ such that $|a_N - \limsup a_i| < \varepsilon$.

Fix $\varepsilon > 0$ and $M \geq 1$. Since $t_n \to \limsup a_i$, choose N_1 such that $|t_n - \limsup a_i| < \varepsilon/2$ for all $n \geq N_1$. Let $n_0 = \max(N_1, M)$, so that $\left|t_{n_0} - \limsup a_i\right| < \varepsilon/2$ and $n_0 \geq M$.

By definition, $t_{n_0} = \sup\{a_m \mid m \geq n_0\}$ is the *least* upper bound of that set. If every a_m with $m \geq n_0$ satisfied $a_m \leq t_{n_0} - \varepsilon/2$, then $t_{n_0} - \varepsilon/2$ would itself be an upper bound smaller than t_{n_0}, contradicting leastness. So there exists some $N \geq n_0 \geq M$ with

$$a_N > t_{n_0} - \varepsilon/2.$$

Also $a_N \leq t_{n_0}$, since t_{n_0} is an upper bound for $\{a_m \mid m \geq n_0\}$ and $N \geq n_0$. Combining,

$$\left|a_N - t_{n_0}\right| < \varepsilon/2.$$

Now use the triangle inequality together with $\left|t_{n_0} - \limsup a_i\right| < \varepsilon/2$:

$$|a_N - \limsup a_i| \leq \left|a_N - t_{n_0}\right| + \left|t_{n_0} - \limsup a_i\right| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Since $N \geq M$, this is exactly what was required. As $\varepsilon > 0$ and $M \geq 1$ were arbitrary, $\limsup a_i$ is a limit point of (a_n). □

Let us now prove Theorem 1.2.

Theorem 1.2

A sequence is cauchy if and only if it is convergent.

Proof. Let a_n be a Cauchy sequence. We claim that it converges to $\limsup a_i$. This exists because of Exercise 1.3.9. and Theorem 1.4.

We wish to find a N such that $|a_n - \limsup a_i| < \varepsilon$ for all $n \geq N$. Since (a_n) is a Cauchy sequence, there exists a N_1 such that $|a_n - a_m| < \varepsilon/4$ for all $n, m \geq N_1$.

Since $t_n := \sup\{a_m \mid m \geq n\}$ is convergent, there exist N_2 such that $|t_n - \limsup a_i| < \varepsilon/2$ for all $n \geq N_2$, and so for $n > N_3 = \max(N_1, N_2)$.

Now consider the terms $a_{N_3}, a_{N_3+1}, \ldots$. Note that t_n is greater than all of them by definition. Now there must exist a i such that $t_n - a_{N_3+i} < \varepsilon/4$, otherwise $t_n - \varepsilon/4$ would be a smaller upper bound of $\{a_m \mid m \geq N_3 + i\}$, which is a contradiction.

Let this be a_N. Since $N > N_3 = \max(N_1, N_2)$, we have that $|a_N - a_m| < \varepsilon/4$ for all $m \geq N$. Using this, and $t_n - a_N = |t_n - a_N| < \varepsilon/4$, we get that:

$$\varepsilon/2 > |a_N - a_m| + |t_n - a_N| \geq |a_m - t_n|$$

and finally, using $|t_n - \limsup a_i| < \varepsilon/2$, we get that:

$$\varepsilon > |t_n - \limsup a_i| + |a_m - t_n| \geq |a_m - \limsup a_i|$$

for all $m \geq N$. Thus we are done. □

Remark 1.4.2. Continuing on Remark 1.2.2, we may also use the extended definition to allow $\limsup a_n$ to be ∞ and $\liminf a_n$ to be $-\infty$. This is a surprisingly important idea, as it allows us to say that *all* sequences, regardless of them being bounded or not, have a lim sup and lim inf.

Remark 1.4.3. Here is a trick I have pulled over you to prove Theorem 1.2. Cauchy sequences are what we use to *construct* the reals. Look carefully at what properties I used for the proof—the most important is the existence of sup. In particular, Cauchy sequences are not all convergent if we work with the rationals! We construct the reals by defining a real to be an equivalence class of Cauchy sequences whose difference is also a Cauchy sequence (i.e the difference between their terms decreases to zero as we grow larger).

Exercises

1.4.10. *Balzano Weirstrass Theorem*

Prove that every bounded sequence has a convergent subsequence.

Hint: You already know one such subsequence! What do subsequences of bounded sequences converge to? But I also want you to do this in another manner, without appealing to what we have already shown, i.e without using the theorems we have already proved for bounded sequences.

1.4.11. Let (a_n) be a bounded sequence. Show that:

1. $\liminf a_n$ exists.
2. $\liminf a_n \leq \limsup a_n$.
3. $\liminf a_n = \limsup a_n$ if and only if (a_n) is convergent (or Cauchy).

Series

2

Solutions to Exercises

Chapter 1

§1.1

Solution 1.1.1.

Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Define

$$T = \{-s : s \in S\}.$$

Since S is bounded below, there exists $m \in \mathbb{R}$ such that $m \leq s$ for all $s \in S$. Then $-s \leq -m$ for all $s \in S$, so T is bounded above by $-m$.

By the least upper bound property of $\mathbb{R}$, T has a supremum; call it $\alpha = \sup T$.

Claim. $-\alpha = \inf S$.

Lower bound. For all $t \in T$, $t \leq \alpha$, so for all $s \in S$, $-s \leq \alpha$, i.e. $-\alpha \leq s$. Thus $-\alpha$ is a lower bound for S.

Greatest lower bound. Let β be any lower bound for S, so $\beta \leq s$ for all $s \in S$. Then $-s \leq -\beta$ for all $s \in S$, i.e. $-\beta$ is an upper bound for T. Since $\alpha = \sup T$ is the *least* upper bound, $\alpha \leq -\beta$, so $\beta \leq -\alpha$.

Hence $-\alpha$ is a lower bound for S and is $\geq$ every other lower bound, so $-\alpha = \inf S$.

§1.2

Solution 1.2.5. **(i) Sum of limits**

Claim.

$$\lim_{n\to\infty}(a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n.$$

Proof. Let $\varepsilon > 0$. Since $a_n \to a$ and $b_n \to b$, choose $N_a, N_b \in \mathbb{N}$ such that

$$n \geq N_a \implies |a_n - a| < \frac{\varepsilon}{2}, \qquad n \geq N_b \implies |b_n - b| < \frac{\varepsilon}{2}.$$

Let $N = \max(N_a, N_b)$. Then for all $n \geq N$,

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| < \varepsilon.$$

$\square$

(ii) Product of limits

Claim.

$$\lim_{n\to\infty}(a_n b_n) = (\lim_{n\to\infty} a_n)(\lim_{n\to\infty} b_n).$$

Proof. Let $\varepsilon > 0$. Choose $N_a, N_b \in \mathbb{N}$ such that

$$\begin{aligned} n \geq N_a &\implies |a_n - a| < \frac{\varepsilon}{2(3|b| + 1)}, \\ n \geq N_b &\implies |b_n - b| < \min\left(\frac{\varepsilon}{2(|a| + 1)}, \frac{|b|}{2}\right). \end{aligned}$$

For $n \geq N_b$, the reverse triangle inequality gives $|b| - |b_n| \leq |b - b_n| < |b|/2$, so $|b_n| < \frac{3}{2}|b|$.

Then for $n \geq N := \max(N_a, N_b)$,

$$\begin{aligned} |a_n b_n - ab| &= |b_n(a_n - a) + a(b_n - b)| \\ &\leq |b_n|\,|a_n - a| + |a||b_n - b| \\ &< \frac{3|b|}{2} \cdot \frac{\varepsilon}{2(3|b| + 1)} + \frac{3|a|}{2} \cdot \frac{\varepsilon}{2(3|a| + 1)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

$\square$

(iii) Quotient of limits

Claim. *If* $\lim_{n\to\infty} a_n = a$ *and* $\lim_{n\to\infty} b_n = b$, *then*

$$\lim_{n\to\infty} \frac{a_n}{b_n} \mathbb{1}_{b_n \neq 0} = \frac{a}{b}.$$

Proof. First, we show that
$$\frac{1}{b_n} \to \frac{1}{b}.$$
Since $b_n \to b$ and $b \neq 0$, there exists $N_1 \in \mathbb{N}$ such that
$$n \geq N_1 \implies |b_n - b| < \frac{|b|}{2}.$$
Hence, for $n \geq N_1$,
$$|b_n| \geq |b| - |b_n - b| > \frac{|b|}{2},$$
so in particular $b_n \neq 0$.

Now let $\varepsilon > 0$. Since $b_n \to b$, choose $N_2 \in \mathbb{N}$ such that
$$n \geq N_2 \implies |b_n - b| < \frac{|b|}{2}.$$
Hence, for $n \geq N_2$,
$$|b_n| \geq |b| - |b_n - b| > \frac{|b|}{2},$$
so in particular $b_n \neq 0$.

Now let $\varepsilon > 0$. Since $b_n \to b$, choose $N_2 \in \mathbb{N}$ such that
$$n \geq N_2 \implies |b_n - b| < \frac{|b|}{2}.$$
Thus, for $n \geq N := \max(N_1, N_2)$,
$$\left| \frac{1}{b_n} - \frac{1}{b} \right| = \frac{|b_n - b|}{|b_n|\,|b|} < \frac{\varepsilon |b|^2/2}{(b/2)|b|} = \varepsilon.$$
Therefore,
$$\frac{1}{b_n} \to \frac{1}{b}.$$
By part (ii), the product of convergent sequences converges to the product of their limits. Hence
$$\frac{a_n}{b_n} = a_n \frac{1}{b_n} \to a \frac{1}{b} = \frac{a}{b}.$$
□

Solution 1.2.6. Suppose that $(a_n)_{n=1}^{\infty}$ is monotonically increasing. Let $a = \sup\{a_n \mid n \in \mathbb{N}\}$. Then,
$$\lim_{n \to \infty} a_n = a.$$

Proof. We proceed by contradiction. Suppose not. Then there exists $\varepsilon > 0$ such that
$$\nexists N \in \mathbb{N} \text{ so that } \forall n > N,\ |a - a_n| < \varepsilon.$$

Note that $a \geq a_n$ for any $n \in \mathbb{N}$ by definition of supremum.

Claim. *For all $K \in \mathbb{N}$ we have that $a - a_K \geq \varepsilon$.*

This is true because we know that there must exist $k^* \geq K$ such that $a - a_{k^*} \geq \varepsilon$. If no such k^* existed, our assumption above would not hold and we would be done.

Now, since $(a_n)_{n=1}^{\infty}$ is monotonically increasing,

$$a_{k^*} \geq a_K \implies a - a_K \geq a - a_{k^*} \geq \varepsilon.$$

Thus, $a - a_K \geq \varepsilon$.

Now, since $\forall K \in \mathbb{N},\ a - a_K \geq \varepsilon$, we have

$$\forall K \in \mathbb{N},\ a - \varepsilon \geq a_K.$$

But then $a - \varepsilon$ is an upper bound of $(a_n)_{n=1}^{\infty}$, which cannot be the case, since $a - \varepsilon < a$, violating the definition of a being a supremum. This is a contradiction, so the claim holds. $\square$

Solution 1.2.7. The sketch of this is as follows. If $(a_n)_{n=1}^{\infty}$ converges to $L \in \mathbb{R}$, then there exist a $N \in \mathbb{N}$ such that $|a_n - L| \leq 1$ for all $n \geq N$. Thus, $|a_n| \leq |L| + 1$ for all $n \geq N$, so $(a_n)_{n=1}^{\infty}$ is bounded by $\max(a_1, a_2, \ldots, a_N, |L| + 1)$

§1.3

Solution 1.3.8. We wish to show that for any $\varepsilon > 0$, $\exists N \in \mathbb{N}$: $\forall n, m \geq N$, $|a_n - a_m| < \varepsilon$.

Since $(a_n)_{n=1}^{\infty}$ is convergent, let $N \in \mathbb{N}$ be such that

$$\forall i > N, \quad |a_i - a| < \frac{\varepsilon}{2}.$$

Then, for any $n, m > N$,

$$|a_n - a| < \frac{\varepsilon}{2}, \qquad |a_m - a| < \frac{\varepsilon}{2}$$

$$\implies |a_n - a_m| = |(a_n - a) + (a - a_m)| \leq |a_n - a| + |a_m - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

$$\implies |a_n - a_m| < \varepsilon \text{ for any } n, m > N.$$

Thus, $(a_n)_{n=1}^{\infty}$ is Cauchy.

Solution 1.3.9. Suppose not. Note that if a sequence is not bounded, then for any $B \in \mathbb{R}$ and any $N \in \mathbb{N}$, we can find $m \in \mathbb{N}$, $m > N$, with $|a_m| > |B|$; if this were not true, the sequence would be bounded above and below by $\max(a_1, \ldots, a_N, B)$ and $\min(a_1, \ldots, a_N, -B)$ respectively.

Now let $(a_n)_{n=1}^{\infty}$ be a Cauchy sequence, and suppose it is not bounded. Since it is Cauchy, for any $\varepsilon > 0$, $\exists K \in \mathbb{N}$ such that

$$|a_m - a_n| < \varepsilon \quad \text{for all } m, n > K.$$

Take $\varepsilon = 1$, and let $K \in \mathbb{N}$ be such that $|a_m - a_n| < 1$ for all $m, n > K$.

Since the sequence is not bounded, there must exist $k^* > K$ such that

$$|a_{k^*}| \geq 1 + |a_K|.$$

But then

$$|a_{k^*} - a_K| \geq |a_{k^*}| - |a_K| \geq 1.$$

However, since $k^*, K > K$, $(a_n)_{n=1}^{\infty}$ being Cauchy gives $|a_{k^*} - a_K| < 1$, a contradiction.

§1.4

Solution 1.4.10.

Claim. *Any sequence (a_n) contains a subsequence which is either nondecreasing (montonically increasing) or non-increasing (monotonically decreasing).*

Proof. Call a natural number n a *peak* point of the sequence (a_n) if $a_m < a_n$ for all $m > n$

Case 1. The sequence has infinitely many peak points. In this case, if $n_1 < n_2 < n_3 < \ldots$ are the peak points, then $a_{n1} > a_{n2} > a_{n3} > \ldots$, so a_{nk} is the desired (nonincreasing) subsequence.

Case 2. The sequence has only finitely many peak points. In this case, let n_1 be greater than all peak points. Since n_1 is not a peak point, there is some $n_2 > n_1$ such that $a_{n2} \geq a_{n1}$. Since n_2 is not a peak point (it is greater than n_1 , and hence greater than all peak points) there is some $n_3 > n_2$ such that $a_{n3} \geq a_{n2}$. Continuing in this way we obtain the desired (nondecreasing) subsequence.

Since we have a monotonically increasing/decreasing sequence, and it must bounded because it is a subsequence of a bounded sequence, by Exercise 1.2.6. we are done. $\square$